

The Neural-Symbolic Query Fabric: Weaving Machine Learning into the Core of Database Operations

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Abstract - Databases manage data with precision but lack learned intelligence, while ML models offer predictive power but operate as external black boxes. The **Neural-Symbolic Query Fabric** integrates compact ML models as first-class citizens within database query engines, enabling native semantic search, anomaly detection, and predictive filtering without costly data movement—transforming databases into **intelligent, proactive data partners**.

1. Problem: The Wall Between Data & Intelligence

Current Architecture: Database (deterministic) + ML (external API, stateless)

Critical Bottlenecks:

- Prohibitive data movement costs (shuffle and movement)
- Unacceptable API latency (high latency)
- Debugging nightmare (deterministic vs. stochastic)

Solution: A database *with* native AI weaving ML directly into query processing as fundamental operators, built on **three pivotal research works**, one of the biggest shifts since the relational model.

2. Pillar 1: In-Database ML

Wang et al. (2024) — VLDB

LLM2Vec: Efficient Vectorization via SLMs

Insight: Distill large models into **<100M parameter SLMs** with **>95% accuracy**—cheap enough for single CPU core.

Implementation: Compact models colocated with data as built-in functions.

SQL Operator:

```
SELECT * FROM docs
WHERE SEMANTIC_SIMILARITY(text, 'query') > 0.85
```

Benefits: Zero latency • Near-zero cost • Seamless SQL

3. Pillar 2: Learned Indexes

Li et al. (2024) — SIGMOD

Axiom: Adaptive Learned Index

Breakthrough: Adaptive structure with **lightweight online learning**—fine-tunes without expensive retraining, maintains B-Tree advantage as workload evolves.

Native Primitive:

```
CREATE INDEX ON sales USING LEARNED INDEX
```

Impact: Access methods become intelligent—modeling complex correlations impossible for B-Trees

4. Pillar 3: Hybrid Optimization

Sharma & Patel (2025) — CIDR

Cascading Optimizer for Neural-Symbolic Queries

Challenge: How does optimizer cost ML models vs. disk I/O?

Solution: Maintain **"neural profiles"** (latency, selectivity, confidence) for intelligent query planning.

Example:

```
SELECT * FROM reviews
WHERE SEMANTIC_SIMILARITY(text, 'positive') > 0.9
AND product_id IN (SELECT id FROM products
WHERE category = 'electronics')
Optimizer runs fast subquery first, applies expensive semantic filter to result.
```

New paradigm: Blend computation cost with I/O cost

5. Use Case: Real-Time Fraud Detection

Traditional: DB → Processor → External ML (Complex & Slow)

DB + Processor + External ML

Neural-Symbolic Fabric
Intelligence Inside Database

Neural-Symbolic Fabric:

Pillar 1: Compact fraud classifier inside DB (microsecond inference)

Pillar 2: Adaptive index on `transaction_amount`, `merchant_id`

Pillar 3: Continuous query as materialized view:

```
CREATE MATERIALIZED VIEW potential_fraud AS
SELECT * FROM transactions
WHERE INFER_FRAUD_RISK(data) > 0.7
AND TRANSACTION_ANOMALY_INDEX(amt, loc) = TRUE
WITH REFRESH FAST ON COMMIT
```

Result: Fraud detection becomes a **native database capability**—faster, simpler, deeply integrated

6. Conclusion

This marks the biggest database shift since the relational model, as intelligence now resides where the data lives.

"Bringing intelligence into the data layer enables data that thinks faster (low-latency), apps that respond smarter (adaptive), and simpler system design."